Discrete Superheterodyne Spectrum Analyzer

Comprehensive Circuit Implementation & LTSpice Simulation Guide

Architecture: Swept Superheterodyne | **RF Range:** 0 to 30 kHz (Audio-to-Ultrasound)
IF Channels: 25 kHz (Ch 2) & 35 kHz (Ch 3) | **Resolution Bandwidth (RBW):** 200 Hz (Q = 125 to 175)
Active Parts: TL072 JFET Op-Amps (10 ICs), 2N2222 / 2N3904 NPN BJTs, 1N4148 Diodes, ADS1115 ADC

1. System Signal Chain Architecture

The instrument follows the classic swept superheterodyne architecture. An incoming RF signal is buffered, mixed down to a fixed intermediate frequency (IF) by a swept local oscillator (LO), filtered through high-Q bandpass stages, envelope detected, smoothed via a video bandwidth (VBW) filter, and displayed against the LO sweep ramp (X-axis):

```
[ RF Input ] --> [ Preamp / Attenuator ] --> [ 2N2222 Gilbert Mixer ] --> [ Active IF Filter Bank ]
|
[ X-OUT (Tuning Ramp) ] <-- [ Sawtooth VCO (LO) ] ------------------------------■
|
v
[ Y-OUT (Spectrum) ] <-- [ Video BW Filter ] <-- [ Envelope Detector ]
```

2. Block-by-Block Circuit Specifications

Block 1: RF Input Preamp & Attenuator (01_RF_Frontend.asc)

- **Function:** High input impedance buffer (1 MΩ) with selectable gain (1x, 10x / +20 dB) to condition weak audio signals (10 mV to 1 Vrms) to optimal mixer level (0.5 to 1.0 Vpp).
- **Components:** TL072 JFET op-amp (±9V to ±12V rails), Cin = 1.0 µF, Rin = 100 Ω, Rbias = 1.0 MΩ, Rf = 9.0 kΩ, Rg = 1.0 kΩ, Cout = 1.0 µF.

Block 2: Discrete Gilbert Cell Four-Quadrant Mixer (02_Gilbert_Mixer.asc)

- **Topology:** 6-transistor active Gilbert cell using 2N2222 BJTs. Driven by single-ended RF and differential LO.
- **Bias Network (+12V rail):** R1=3.3k, R2=2.2k, R4=2.2k, R3=1.5k; Base resistors R8..R11, R6 = 10 kΩ; Tail resistor R5 = 390 Ω; Collector loads R12=R7=2.2 kΩ.
- **Coupling Capacitors:** Bypass C1, C2, C3 = 10 µF electrolytic; LO coupling C4, C5 = 100 nF; Output Cout = 1.0 µF.

Block 3: IF Active Bandpass Filter Bank (03_IF_Filter_Bank.asc)

- **Channel 2 (25 kHz Sallen-Key):** f0 = 25,000 Hz, BW = 200 Hz (Q = 125). TL072 op-amps, R3=R5=796 kΩ (1%), R4=6.34 kΩ (1%), C3=C4=1.0 nF (C0G). Includes input divider (10k / 2.2k, -15 dB) to prevent passband clipping.
- **Channel 3 (35 kHz MFB Filter Redesign):** f0 = 35,000 Hz, BW = 200 Hz (Q = 175, Av = 2.0). Replaced oscillating Sallen-Key with Multiple Feedback (MFB) topology. Components: C1=C2=1.0 nF, R3 (feedback) = 1.591 MΩ, R1 (input) = 397.8 kΩ, R2 (ground) = 12.98 Ω.

Block 4: Diode Envelope Detector & Buffer (04_Envelope_Detector.asc)

- **Components:** 1N4148 fast rectifier diode, C = 10 nF reservoir cap, R = 22 kΩ discharge resistor (tau = 220 µs). TL072 unity-gain buffer.

Block 5: Video Bandwidth (VBW) Low-Pass Filter (05_VBW_Filter.asc)

- **Selectable Modes:** Low (fc = 50 Hz, R = 33k, C = 100n), Medium (fc = 200 Hz, R = 10k, C = 82n), High (fc = 1 kHz, R = 10k, C = 15n).

Block 6: Sawtooth Sweep Generator & VCO (06_Sawtooth_VCO.asc)

- **Topology:** TL072 Integrator (Ctime = 10n..100n, Rcharge = 100k, Icharge = 50 µA) + TL072 Schmitt Trigger + 2N3904 Reset BJT. Produces a 0V to 5V linear ramp tuning LO from 20 kHz to 50 kHz.

3. Comprehensive Simulation Suite (Implementation & Directives)

Sim 1: AC Frequency Response & Stability Analysis

Goal & Description: Validates resonant frequency f0 = 25 kHz ± 100 Hz, BW = 200 Hz ± 25 Hz, and verifies unconditional stability (monotonic phase decrease).

```
.ac dec 200 10k 50k
.meas AC f_center MAX mag(V(IF_out_2))
.meas AC f_low WHEN mag(V(IF_out_2))=f_center/sqrt(2) FALL=1
.meas AC f_high WHEN mag(V(IF_out_2))=f_center/sqrt(2) RISE=1
.meas AC bw PARAM (f_high - f_low)
```

Sim 2: Component Tolerance & Monte Carlo Analysis

Goal & Description: 100-run statistical sweep ensuring >95% of runs remain within 25 kHz ± 250 Hz (1.0%) center frequency spread with 1% resistors and 5% caps.

```
.param R3_val = {mc(796k, 0.01)}
.param R4_val = {mc(6.34k, 0.01)}
.param C3_val = {mc(1n, 0.05)}
.param C4_val = {mc(1n, 0.05)}
.step param run 1 100 1
```

Sim 3: Worst-Case Sensitivity Analysis

Goal & Description: Evaluates maximum frequency error bounding box (delta_f_max) across all binary tolerance corner permutations.

```
.param R3_wc = {wc(796k, 0.01, 1)}
.param R5_wc = {wc(796k, 0.01, 2)}
.param C3_wc = {wc(1n, 0.05, 3)}
.param C4_wc = {wc(1n, 0.05, 4)}
.step param run 1 16 1
```

Sim 4: Thermal Drift Analysis

Goal & Description: Quantifies frequency shift across temperature caused by semiconductor Vbe tempco (-2 mV/°C), op-amp offset drift, and resistor TCR (< 50 Hz drift).

```
.model RES_PRECISION R (tc1=50e-6) ; 50 ppm/°C metal film
.step temp 0 70 10
```

Sim 5: Mixer Intermodulation Distortion & Linearity (FFT)

Goal & Description: FFT on V(mix_out). Measures mixing products (25.0, 25.2 kHz) and 3rd-order intermod products (24.8, 25.4 kHz). Verifies SFDR >= 45 dBc.

```
* Two-Tone Input: 5.0 kHz + 5.2 kHz (100 mVpk each); LO = 20.0 kHz
.tran 0 20m 10m 100n
```

Sim 6: Master Dynamic Multi-Tone Sweep

Goal & Description: Plots V(env_out) vs V(ramp) in simulated X-Y display mode. Confirms 3 distinct, symmetric spectral peaks without asymmetrical ringing tails.

```
* Three-tone RF: 4.0 kHz + 7.0 kHz + 12.0 kHz; Sweeping LO: 20 kHz to 40 kHz
.tran 0 100m 0 1u
```

4. Summary of Simulation Deliverables

#	Schematic File	Simulation Type	Key Directive	Target Deliverable
1	03_IF_Filter_Bank.asc	AC Response	.ac dec 200 10k 50k	f0 = 25 kHz, BW = 200 Hz, Q = 125, Stable
2	03_IF_Filter_Bank.asc	Monte Carlo	.step param run 1 100 1	Δf0 < ±250 Hz across 100 runs (1% R)
3	03_IF_Filter_Bank.asc	Temperature	.step temp 0 70 10	Δf0 < 50 Hz over ΔT = 40 °C
4	02_Gilbert_Mixer.asc	Linearity / FFT	.tran 0 20m 10m 100n	SFDR ≥ 45 dBc, conversion gain
5	06_Sawtooth_VCO.asc	Ramp Linearity	.tran 0 50m 0 1u	0V to 5V linear ramp, flyback < 0.5 ms
6	SpectrumAnalyzer_Master .asc	Master Sweep	.tran 0 100m 0 1u	Full 3-tone spectrum in X-Y display mode